

A classical star family and the affine- D_4 boundary

Explicit proof and independent computational certificate

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Abstract

A broad search rediscovered a simple fully commutative-bottom family with Catalan leading coefficients. Its four-leaf case is affine D_4 and has $P_{x,w} = 1 + 3q + 2q^2$. We give an elementary proof and an independent affine-action computation. This supplies a precise obstruction to extending the type- D fully commutative-bottom bound to every Coxeter system. The exact star family is explicitly covered by Mongelli's 2011 preprint, Corollary 2, as well as older cube-interval formulas; it is not presented as a new discovery or as a use requiring the September 2026 invariance theorem.

1 The star Coxeter systems

Let W_m have generators $c, a_1, \dots, a_m$, with $m(c, a_i) = 3$ and $m(a_i, a_j) = 2$ for $i \neq j$. For $L \subseteq [m]$ put $a_L = \prod_{i \in L} a_i$ and

$$x = a_{[m]}, \quad w = xc x.$$

The lower endpoint is a product of distinct commuting generators, hence fully commutative. The systems for $m = 1, 2, 3, 4$ have types $A_2, A_3, D_4, \tilde{D}_4$; for $m \geq 5$ the form is indefinite.

Theorem 1 (Star family; classical polynomial formula). *The endpoints have lengths m and $2m + 1$, and*

$$P_{x,w}(q) = \sum_{j=0}^{\lfloor m/2 \rfloor} \left(\binom{m}{j} - \binom{m}{j-1} \right) q^j, \quad \binom{m}{-1} = 0.$$

For $m = 2k$ its leading μ -coefficient is

$$\mu(x, w) = \binom{2k}{k} - \binom{2k}{k-1} = \frac{1}{k+1} \binom{2k}{k}.$$

For odd m the length difference is even and $\mu(x, w) = 0$.

The coefficient sequence is unbounded as k grows. This assertion concerns the changing star systems; only $m = 4$ is the affine- D_4 case.

2 The complete interval, not just its rank counts

Proof of Theorem 1. Every $a_L c a_R$ is reduced of length $|L| + |R| + 1$. To see this, use the faithful geometric representation with simple roots α_c, α_i . The initial commuting letters and the subsequent c multiply at ascents. After c , each as-yet unused right letter a_j sees the positive root

$$\alpha_c + \sum_{i \in L} \alpha_i + (-1)^{[j \in L]} \alpha_j.$$

Right multiplication by the other leaves does not change that column. Thus each right letter is also an ascent.

More generally the image of α_j under $a_L c a_R$ is

$$(-1)^{[j \in R]} \left(\alpha_c + \sum_{i \in L} \alpha_i + (-1)^{[j \in L]} \alpha_j \right).$$

Its center coefficient determines whether $j \in R$; its diagonal leaf coefficient is zero precisely when $j \in L$. Consequently all pairs (L, R) give distinct elements.

The fixed reduced word for w and the subword criterion now give every element below it: either a commuting leaf product or an $a_L c a_R$. The elements at or above x are x itself and exactly those $a_L c a_R$ with $L \cup R = [m]$. For such pairs the order relation is

$$a_L c a_R \leq a_{L'} c a_{R'} \iff L \subseteq L', \quad R \subseteq R'.$$

Indeed, a reduced subword producing an element with c -support must retain c , and the distinct-pair formula prevents another representation from changing either subset. A subword omitting c gives x exactly when all leaves are selected once.

Each coordinate is therefore left-only, right-only, or both. Order permits either of the first two states to move to “both”. With x as the empty face, this is the entire face lattice of an m -cube. There are $3^m + 1$ elements. Relative rank $r + 1$ contains $\binom{m}{r} 2^{m-r}$ elements, for $0 \leq r \leq m$.

For a nonempty face y , every interval $[y, z]$ is Boolean, so $R_{y,z}(q) = (q-1)^{\ell(z)-\ell(y)}$ and $P_{y,z}(q) = 1$. For the bottom x and a face $z = a_L c a_R$, we also have

$$R_{x,z}(q) = (q-1)^{|L \cap R|+1}.$$

This follows directly from the ordinary R recurrence: cancel common right descents a_R from both endpoints, and then cancel common left descents $a_{[m] \setminus R}$ from both endpoints. The resulting pair is $(e, a_{L \cap R} c)$, a word with no repeated generator.

Put $F = P_{x,w}$. Ordinary KL reciprocity, using all the nonempty faces, therefore becomes

$$\begin{aligned} q^{m+1} F(q^{-1}) - F(q) &= \sum_{r=0}^m \binom{m}{r} 2^{m-r} (q-1)^{r+1} \\ &= (q-1)(q+1)^m. \end{aligned}$$

The ordinary degree bound gives $\deg F \leq \lfloor m/2 \rfloor$. In that degree range the coefficient on the left is $-[q^j]F$. The coefficient on the right is $\binom{m}{j-1} - \binom{m}{j}$, proving the formula. The asserted μ -values follow from the interval rank $m+1$. $\square$

The Boolean-interval formulas and related lattice formulas are classical [1]. The top xcx is a boolean reflection: because the leaves commute it is the palindromic word $a_1 \cdots a_m c a_m \cdots a_1$. Both endpoints therefore belong to the older tree-graph boolean-element framework of Mongelli. His Corollary 2 explicitly displays the star pattern and its Catalan μ -coefficient; his displayed f_4 polynomial is $1 + 3q + 2q^2$ [2, preprint pp.6 and 10]. The whole proposed family is therefore prior art.

3 An independently checked affine example

Use affine- D_4 labels with central node 2 and leaves 0, 1, 3, 4. The concrete pair is

$$x = s_0 s_1 s_3 s_4, \quad w = x s_2 x, \quad P_{x,w} = 1 + 3q + 2q^2, \quad \mu(x, w) = 2.$$

An independent verifier uses affine signed permutation actions on $\mathbb{R}^4$. Generator 0 is the signed swap of the first two coordinates, generators 1, 2, 3 are adjacent swaps, and generator 4 is the signed swap of the last two coordinates followed by translation by $e_3 + e_4$. These give the specified Coxeter diagram.

The verifier uses direct subword sets for Bruhat comparison and computes the polynomials through R -reciprocity rather than the discovery program's KL descent recurrence. It checks all 271 nontrivial reciprocity identities in the 272-element principal lower ideal. The closed interval has 82 elements with rank counts

$$(1, 16, 32, 24, 8, 1).$$

The faithful affine actions of the two endpoints, represented as signed coordinate images followed by translation, are

$$x = ((-1, -2, -3, -4), (0, 0, 1, 1)), \quad w = ((1, 3, 2, 4), (0, -1, 1, 0)).$$

All code is saved. Run

```
python3research/verify_affine_d4_r.py
Certificate: results/affine-d4-independent.json
Discovery engine: research/broad_cells/targeted.py
```

4 What this establishes, and what it does not

The affine example disproves the universal statement that a fully commutative lower endpoint forces $\mu \in \{0, 1\}$ in every Coxeter group. Green's introduction describes a possible universal scope rather than a numbered universal conjecture [3, printed pp.2–3]; the distinction matters when describing this boundary. We do not claim to be the first to observe its failure: the explicit polynomial family is in Mongelli's 2011 preprint. The priority report records the exact matching star diagram.

This example does not refute Gern's finite type- D theorem [4, Theorem 4.5.11] or the companion finite- E_6/E_7 theorem. It also does not, by itself, give a counterexample in every higher affine type D : the degree-four star is not a standard parabolic diagram in those systems. The family neither requires new combinatorial invariance nor supplies the sought positive unbounded finite- D model family.

References

- [1] F. Brenti, A combinatorial formula for Kazhdan–Lusztig polynomials, *Invent. Math.* 118 (1994), 371–394, Corollaries 6.8–6.9. [Primary extended abstract](#).
- [2] P. Mongelli, Kazhdan–Lusztig polynomials of boolean elements, *J. Algebraic Combin.* 39 (2014), 497–525. [arXiv:1111.2945](#).
- [3] R. M. Green, Leading coefficients of Kazhdan–Lusztig polynomials and fully commutative elements, *J. Algebraic Combin.* 30 (2009), 165–171. [arXiv:0801.1650](#).
- [4] T. C. Gern, Leading coefficients of Kazhdan–Lusztig polynomials in type D , Ph.D. thesis (2013). [arXiv:1304.6074](#).